Code No: 124CQ **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
DATABASE MANAGEMENT SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Write short notes on DDL and DML [2]
b) List various symbols used in ER diagrams. [3]
c) "Relational algebra is procedural and tuple relational calculus is non- procedural language". Justify. [2]
d) Write a note on correlated sub queries. [3]
e) Explain about multivalued dependency. [2]
f) What are the problems caused by redundancy? [3]
g) What do you mean by serializability of a schedule? [2]
h) Explain different states of transaction. [3]
i) What is local depth? [2]
j) What is primary index? [3]

PART-B

(50 Marks)

2.a) With a neat diagram, explain the components of database system structure.
b) Draw an ER diagram for hospital management database. [5+5]

OR

3.a) Explain the process of translating ER diagrams to relations.
b) Describe three level architecture of database schema. [5+5]

4.a) Consider the following relations.
Emp(Ename, Street, City)
Works(Ename, Cname, Salary)
Company(Cname, City)
Manages(Ename, Mname)
Express the following queries in relational algebra.
i) Find names of all employees who earn more than every employee of "Bank Corporation".
ii) List the city of leaving of all the managers.
b) Define Trigger. Differentiate row level and statement triggers. [5+5]

OR

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5.a) Differentiate tuple and domain relational calculus.

b) Consider the following relations

Suppier(Sid, Sname, Address)

Parts(Pid, Pname, Color)

Catalog(Sid, Pid, Cost)

Write the following queries in SQL

i) Find the Snames of the suppliers who supply every part.

ii) For each part, find the Sname of the supplier who charges the most for that part. [5+5]

6.a) Write the procedure to find minimal cover of functional dependency set.

b) Define Normal form. Explain 2NF, 3NF and BCNF with examples. [5+5]

OR

7.a) What are the properties of decomposition? Explain in detail.

b) Explain the following terms:

i) Functional dependency.

ii) Partial Functional Dependency

iii) Transitive Dependency. [4+6]

8.a) Explain the need for locking and discuss two phase locking protocol in detail.

b) What is shadow paging? Explain. [5+5]

OR

9.a) Explain about multiple granularity.

b) Discuss about recoverable schedules and cascadeless schedules. [5+5]

10.a) Explain the process of inserting a record in to a B^+ tree.

b) Explain about linear hashing. [5+5]

OR

11.a) What is the role of global depth in extendible hashing?

b) Differentiate clustered and un-clustered index. [5+5]

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